Big-O and function composition

Asked 8 years, 5 months ago Modified 10 months ago Viewed 7k times

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On the last page of this document, a property of big-O operations is listed which says that if

$$f_1(n) = O(g_1(n)), \text{ and } f_2(n) = O(g_2(n))$$

Then

$$f_1 \circ f_2 = O(g_1 \circ g_2)$$

Why is that?

Trying to get this to a form that resembles the definition I get to:

$$f_1(f_2(n)) = k_1 * g_1(g_2(n))$$

and I'm kind of stuck... I don't see how the big-O definition inequality holds.

UPDATE: We can assume the functions are increasing monotonically.

functions asymptotics computational-complexity function-and-relation-composition

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edited Nov 29, 2021 at 3:43
u243676
105 4

asked Apr 19, 2014 at 21:30
Rire1979
399 5 9

2 Answers

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This is false unless you make extra assumptions. Assume that $f_2(x) = 2x$, $g_2(x) = x$. Then $f_2(x) = O(g_2(x))$. Now take $f_1(x) = g_1(x) = e^x$. Then you get $f_1(f_2(x)) = e^{2x}$ and $g_1(g_2(x)) = e^x$, and $e^{2x} \neq O(e^x)$ so there's something wrong.

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answered Apr 19, 2014 at 21:40
user2566092
25.8k 29 61

"$f = O(g)$" means that for some c and some x_0, $f(x) \leq c \cdot g(x)$ for all $x \geq x_0$.

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So, if $f_1 = O(g_1)$ and $f_2 = O(g_2)$, there are c_1, c_2 and an x_0 with

$$f_1(x) \leq c_1 g(1), \quad f_2(x) \leq c_2 g_2(x) \text{ for all } x > x_0.$$

So what, then, can we say about $f_1 \circ f_2$?. We have for every $x > x_0$ that

$$y = f_2(x) \leq c_2 g_2(x)$$

and therefore

$$(f_1 \circ f_2)(x) = f_1(y) \leq c_1 g_1(y).$$

If we additionally assume that g increases monotonically, then we can replace y with the *larger value* $c_2 g_2(x)$ in that last inequality to get

$$(f_1 \circ f_2)(x) = f_1(y) \leq c_1 g_1(c_2 g_2(x)).$$

which implies that

$$f_1 \circ f_2 = O(g_1 \circ c_2 g_2).$$

user2566092's example shows that we cannot, in general, get rid of that factor c_2 there. Though if, for example, $g_1(x) = x^n$, then
$g_1(c_2 g_2(x)) = (c_2 g_2(x))^n = c_2^n (g_2(x))^n = O((g_1)^n)$, so for *these* g_1 you indeed have

$$f_1 \circ f_2 = O(g_1 \circ g_2).$$

The general property that you need for your proposition to hold is thus that

$$g_1(c \cdot x) = O(g_1(x)).$$

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edited Apr 20, 2014 at 0:13

answered Apr 19, 2014 at 21:50



fgp
20.2k 2 30 48

I thought big-O was for some $x_0 \in \mathbb{R}, \forall x \geq x_0, \ldots$ – ObjectsMorphisms Apr 19, 2014 at 22:18

For x, where x > xo – Rire1979 Apr 19, 2014 at 22:26

@EnjoysMath Hm, true, I was sloppy there. The conclusions still hold, though, once just needs to replace "forall x" by "forall $x > x_0$". – fgp Apr 19, 2014 at 22:29